

Online Payment Fraud Detection Using Machine Learning

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CHAPTER 1

INTRODUCTION

1.1 Project Overview

Online payment systems have become an essential part of digital transactions due to the rapid growth of e-commerce, mobile banking, and digital wallets. However, the increasing use of online transactions has also led to a significant rise in fraudulent activities such as unauthorized transactions, identity theft, and financial scams.

The proposed project, **Online Payment Fraud Detection Using Machine Learning**, focuses on identifying fraudulent transactions in real-time by analyzing transaction patterns and behavioral features. Machine learning algorithms are used to study historical transaction data and detect suspicious activities automatically.

The system processes transaction details such as transaction amount, type, time, and user behavior, and predicts whether a transaction is **legitimate or fraudulent**. By applying data preprocessing, feature selection, model training, and evaluation techniques, the system ensures accurate and efficient fraud detection.

This project helps financial institutions and online platforms reduce financial losses, improve security, and enhance customer trust by preventing fraudulent activities before they cause damage.

1.2 Purpose of the Project

The main purpose of this project is to design and develop a machine learning-based system that can automatically detect fraudulent online payment transactions with high accuracy.

The specific objectives of the project are:

- To analyze online payment transaction data and identify fraud patterns.
- To implement and compare multiple machine learning algorithms for fraud detection.
- To improve detection accuracy through data preprocessing and hyperparameter tuning.
- To minimize false positives (legitimate transactions marked as fraud).
- To build a system capable of real-time fraud prediction.
- To enhance digital payment security and reduce financial risks.

The system aims to support banks, financial institutions, and e-commerce platforms in preventing fraud and ensuring safe digital transactions.

CHAPTER 2

IDEATION PHASE

2.1 Problem Statement:

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - who are they?	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - what are they trying to achieve?	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way - what bothers them most?	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists - what needs to be solved?	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view - how does it impact them emotionally?	Describe the emotions the result from experiencing the problems or barriers

Reference: <https://miro.com/templates/customer-problem-statement/>

Example:

I am a traveler	I'm trying to book flights on my phone	but it takes a long time	Because The website is not responsive and doesn't have a mobile version	which makes me feel Frustrated
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Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A digital payment user who frequently uses UPI/credit/debit cards for online transactions	To complete secure and fast online payments	I am worried about fraud and unauthorized transactions	Fraudsters use advanced techniques and current systems may not detect fraud instantly	Anxious, insecure, and stressed about losing money
PS-2	A bank or financial service provider handling thousands of daily transactions	To prevent fraudulent transactions while ensuring smooth customer experience	It is difficult to detect fraud in real-time without blocking legitimate users	Fraud patterns constantly evolve and datasets are highly imbalanced	Pressured, concerned about financial loss and customer trust

2.2 Empathy Map Canvas:

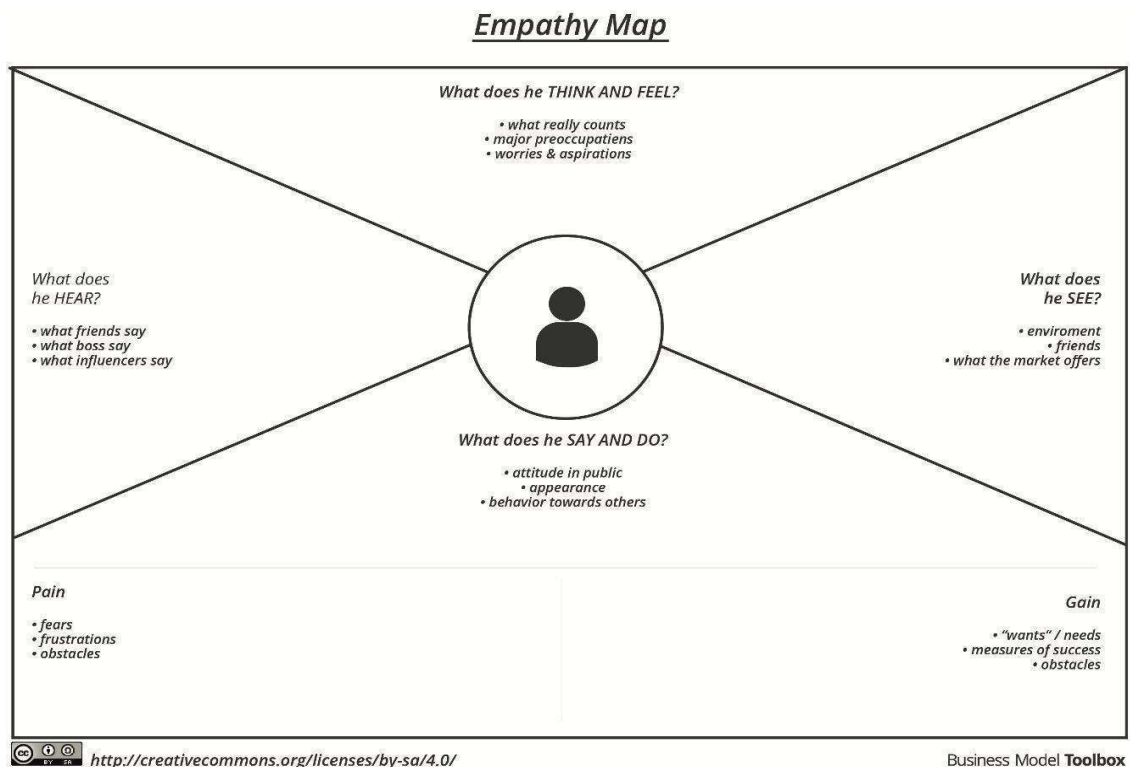
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

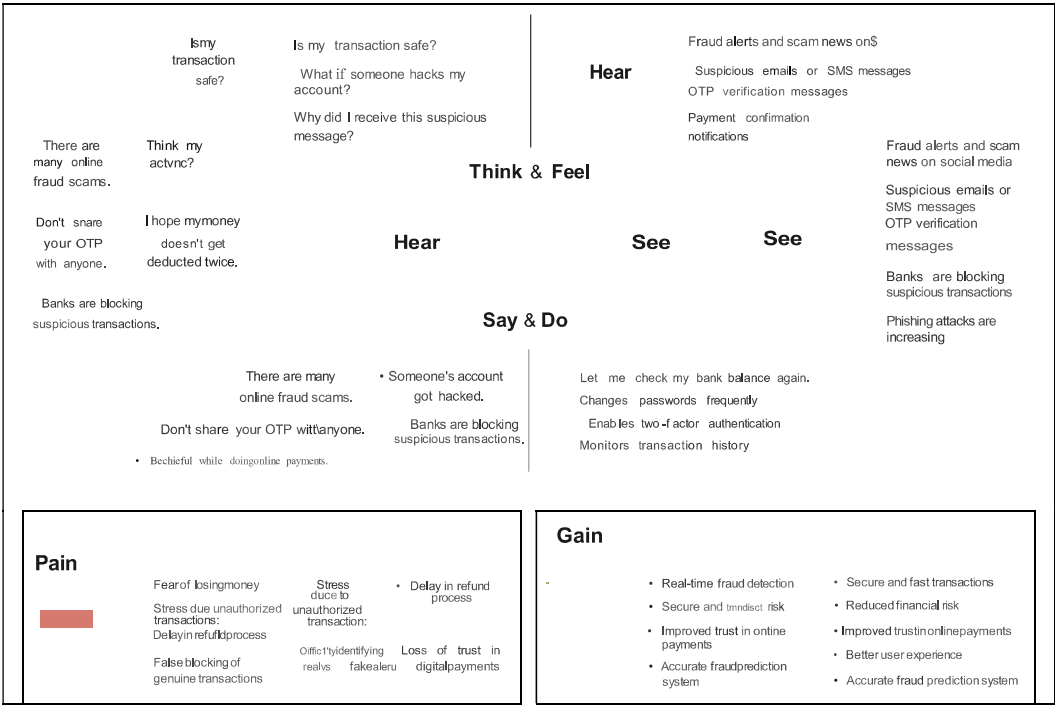
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Reference: <https://www.mural.co/templates/empathy-map-canvas>

Example: Online Payment Fraud Detection



2.3 Brainstorm & Idea Prioritization Template:

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

🕒 10 minutes to prepare
🕒 1 hour to collaborate
👤 2-8 people recommended

➔

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

🕒 10 minutes

A Team gathering
Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

B Set the goal
Think about the problem you'll be focusing on solving in the brainstorming session.

C Learn how to use the facilitation tools
Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#) ➔

1

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

🕒 5 minutes

PROBLEM

How might we [your problem statement]?

Key rules of brainstorming

To run a smooth and productive session

- 🗣️ Stay in topic.
- 💡 Encourage wild ideas.
- 👂 Defer judgment.
- 👂 Listen to others.
- 🗣️ Go for volume.
- 👁️ If possible, be visual.

Brainstorm

Write down any ideas that come to mind that address the problem of detecting online payment fraud.

(t) 10 minutes

Use transaction amount analysis to detect unusual spending patterns

Analyze transaction frequency per user

Detect sudden large transactions

Compare domestic vs international transactions

Analyze device ID & IP address patterns

Detect multiple failed login attempts

Use heatmaps to check feature correlation

Compare fraud vs non-fraud transaction distribution

Group ideas

Take turns sharing ideas while clustering similar or related notes together. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

(t) 20 minutes

Transaction Behavior Analysis

Multiple Failed Login Attempts

Fraud vs Non-Fraud Distribution

Machine Learning Model Development

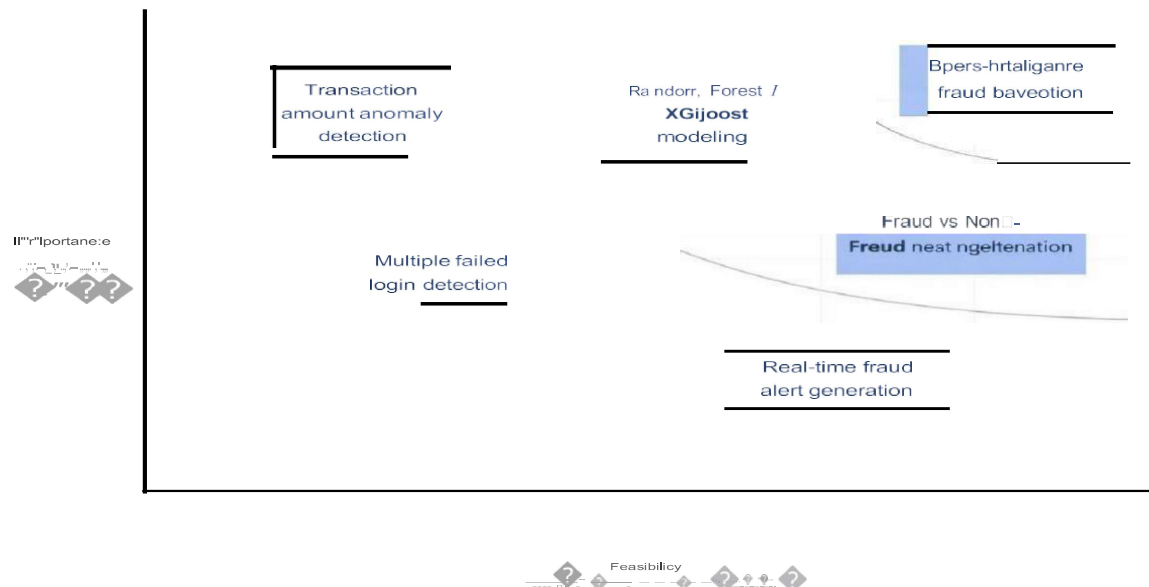


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Prioritize

Your team should all be on the same page about what's important, moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

C) 20 minutes



CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Registration through Email & Password Secure Login with Username & Password Two-Factor Authentication (OTP Verification)
FR-2	Transaction Processing	Capture transaction details (amount, location, device info) Validate transaction input data Store transaction in database
FR-3	Fraud Detection	Extract transaction features for analysis Apply trained ML model for fraud prediction Classify transaction as Fraud / Legitimate
FR-4	Alert & Notification System	Send real-time fraud alert to user Allow user to confirm or reject flagged transaction
FR-5	Fraud Reporting & Logging	Store prediction results in fraud logs Maintain transaction history
FR-6	Admin Dashboard	Monitor fraud statistics (accuracy, precision, recall)
Fr-7	Model Management	Retrain model using updated dataset Compare model performance metrics Update deployed model

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have a simple and user-friendly interface for customers and administrators.
NFR-2	Security	All transaction data must be encrypted. OTP-based authentication and secure APIs must be implemented.
NFR-3	Reliability	The fraud detection system must provide accurate and consistent predictions with minimal false positives.
NFR-4	Performance	Fraud prediction must be generated within milliseconds to support real-time transaction processing.
NFR-5	Availability	The system should be available 24/7 with minimal downtime.
NFR-6	Scalability	The system should handle increasing transaction volume without performance degradation.
NFR-7	Maintainability	The model and system should allow easy updates and retraining.
NFR-8	Auditability	All transactions and fraud decisions must be logged for auditing and review purposes.

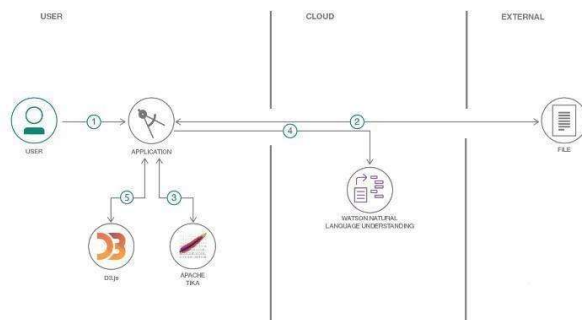
3.2 Data Flow Diagram & User Stories

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

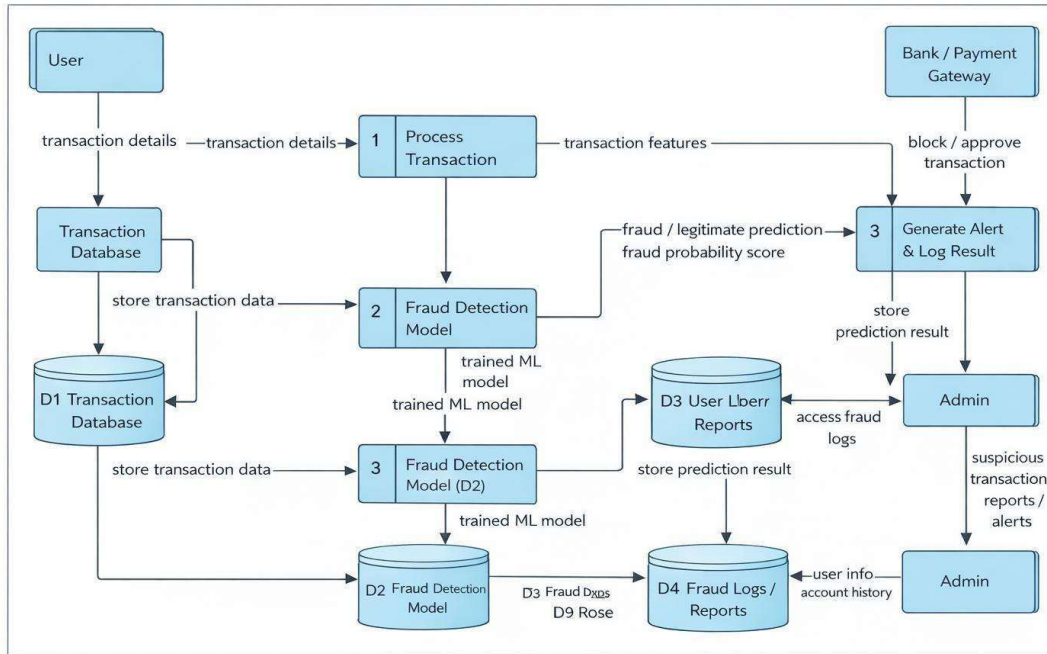
Example: [\(Simplified\)](#)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD Level 0 - Online Payment Fraud Detection System



User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer	Fraud Monitoring	US-1	As a customer, I want to receive real-time alerts when a high-risk transaction occurs on my account.	System sends instant notification (SMS/App/Email) for risky transactions.	High	Sprint-1
Customer	Transaction Transparency	US-2	As a customer, I want to view the risk score of my recent transactions.	User can see fraud probability score and transaction status.	High	Sprint-2
Customer	Account Security	US-3	As a customer, I want the system to temporarily block suspicious	Suspicious transaction is placed on hold until user confirmation.	High	Sprint-1

			transactions until I verify them.			
Customer	Authentication	US-4	As a customer, I want secure login using two-factor authentication.	User logs in using password + OTP verification.	High	Sprint-1
Fraud Analyst	Monitoring	US-5	As a fraud analyst, I want to review flagged transactions in a centralized dashboard.	Analyst can filter transactions by risk level, date, and user.	High	Sprint-2
Fraud Analyst	Investigation	US-6	As a fraud analyst, I want to see detailed transaction patterns for suspicious accounts.	System displays transaction history and anomaly indicators.	High	Sprint-3
Administrator	Reporting & Analytics	US-7	As an admin, I want to generate monthly fraud detection performance reports.	Admin can download reports showing precision, recall, F1-score, and accuracy.	High	Sprint-3

3.3 Technology Stack (Architecture & Stack):

Technical Architecture:

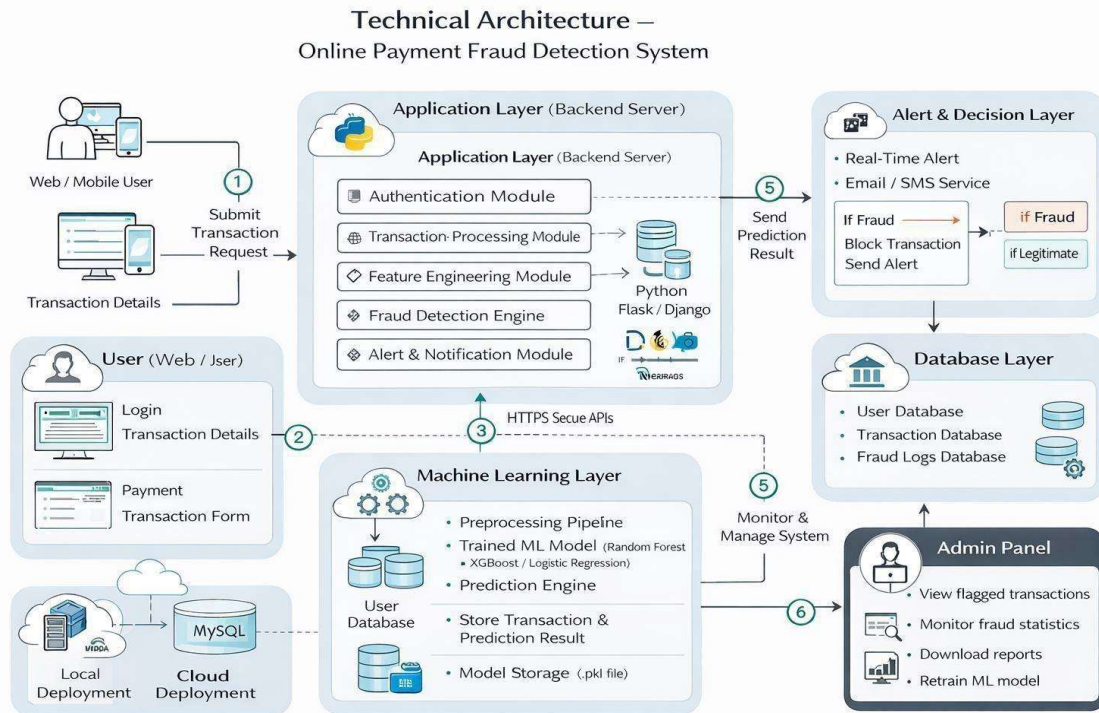


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web-based interface for users and admin to perform transactions and view fraud alerts	HTML, CSS, JavaScript, Bootstrap / React JS
2	Application Logic-1	Transaction processing and validation logic	Python (Flask / Django)
3	Application Logic-2	Fraud detection logic using trained ML model	Python (Scikit-learn, Pandas, NumPy)
4	Database	Store user details, transaction history, fraud logs	MySQL / PostgreSQL

5	Cloud Database	Cloud-based storage for scalable transaction data	AWS RDS / MongoDB Atlas
6	File Storage	Storage of datasets, trained ML model (.pkl), reports	Local File System / AWS S3
7	External API-1	Payment gateway integration for transaction validation	Razorpay API / Stripe API
8	Machine Learning Model	Predict fraudulent vs legitimate transactions	Random Forest / Logistic Regression / XGBoost
9	Infrastructure (Server / Cloud)	Application deployment environment	Local Server (Windows/Linux), AWS EC2 / Heroku

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for backend, ML, and frontend	Flask, Scikit-learn, Pandas, NumPy, Bootstrap
2	Security Implementations	Encryption, authentication, secure APIs, input validation	HTTPS, SHA-256 Password Hashing, JWT Authentication, OTP Verification
3	Scalable Architecture	3-Tier Architecture (UI → Application → Database) supporting future microservices scaling	Flask (API Layer), MySQL, AWS EC2
4	Availability	System available 24/7 with minimal downtime using cloud deployment	AWS EC2 / Load Balancer
5	Performance	Real-time fraud detection with millisecond-level prediction, optimized queries	Scikit-learn optimized model, Indexed Database, Caching
6	Data Privacy	Secure handling of user financial data	Data Encryption, Secure APIs, Access Control
7	Maintainability	Easy model retraining and system updates	Modular Python Code, Version Control (Git)

CHAPTER 4

PROJECT DESIGN

4.1 Problem Solution Fit Template:

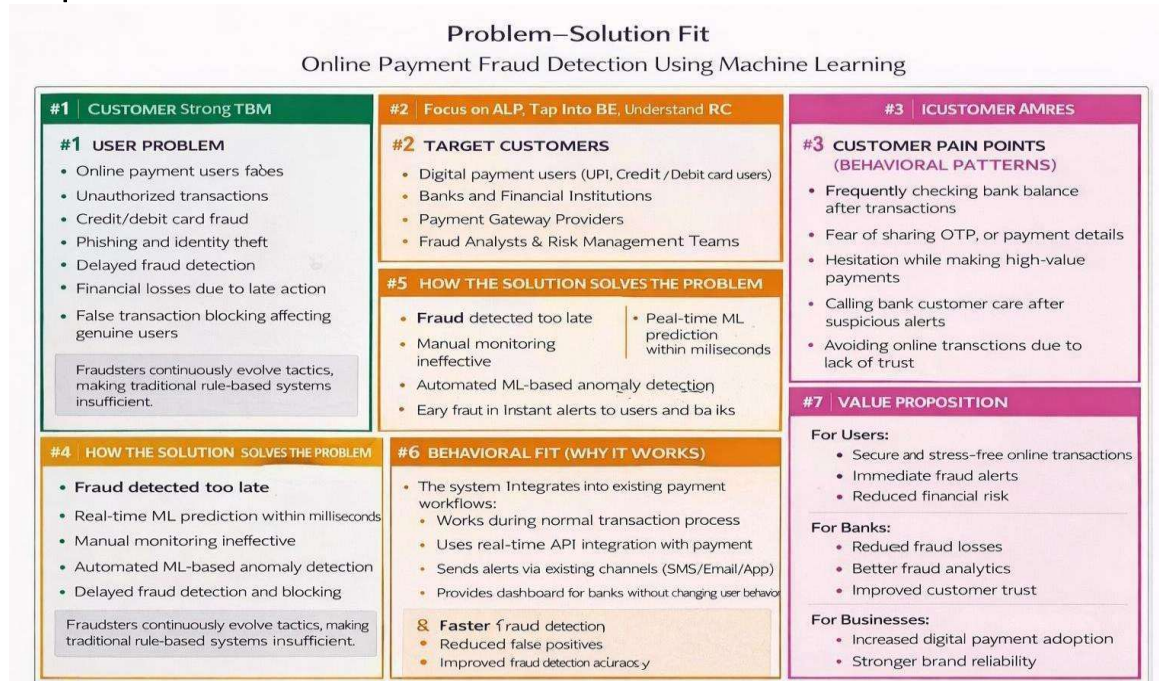
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- Understand the existing situation in order to improve it for your target group.

Template:



4.2 Proposed Solution:

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Online payment users and financial institutions face increasing risks of fraudulent transactions, identity theft, phishing attacks, and delayed fraud detection. Traditional rule-based systems fail to detect complex and evolving fraud patterns in real-time, leading to financial losses and reduced trust in digital payments.
2	Idea / Solution Description	Develop a Machine Learning-based Online Payment Fraud Detection System that analyzes transaction data in real-time, generates fraud probability scores, and classifies transactions as Fraud or Legitimate. The system sends instant alerts, blocks high-risk transactions, logs activities, and provides an admin dashboard for monitoring fraud statistics and model performance.
3	Novelty / Uniqueness	Real-time fraud prediction using ML models (Random Forest/XGBoost). Fraud probability scoring instead of simple rule-based detection. Automated alert & blocking mechanism. Integration-ready API for banks/payment gateways. Continuous model retraining capability for adapting to new fraud patterns.
4	Social Impact / Customer Satisfaction	Reduces financial fraud and cybercrime impact. Increases trust in digital payments. Enhances financial security for individuals and businesses. Protects vulnerable users from phishing and scams. Improves customer confidence and satisfaction.

5	Business Model (Revenue Model)	Subscription-based model for banks and payment gateways. API-based pricing per transaction analyzed. SaaS (Software as a Service) model deployment. Enterprise fraud analytics dashboard licensing.
6	Scalability of the Solution	Cloud deployment (AWS/Azure) for handling high transaction volumes. Microservices-ready architecture. Model retraining with large datasets. Scalable database for millions of transactions. API-based integration for multiple banks and fintech platforms.

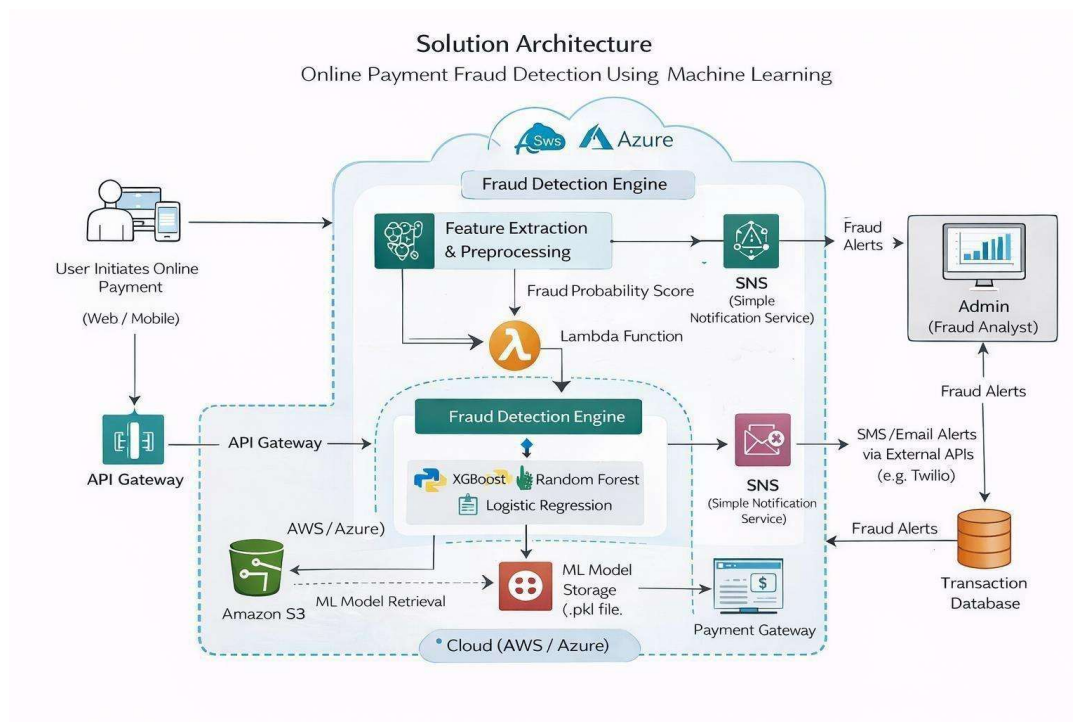
4.3 Solution Architecture

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Online Payment Fraud Detection using Machine Learning



CHAPTER 5

PROJECT PLANNING & SCHEDULING

5.1 Project Planning:

Product Backlog, Sprint Schedule, and Estimation

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member
Sprint-1	User Registration	USN-1	As a user, I want to register using email and password, so that I can access the system securely.	2	High	Sneha Alishetty
Sprint-1	User Registration	USN-2	As a user, I want to receive an email confirmation after registration, so that my account is verified.	1	High	Somisetty Vasanth Kumar
Sprint-1	Login	USN-3	As a user, I want to log in using my email and password, so that I can access my dashboard.	1	High	Syed Aashifa
Sprint-2	Fraud Detection	USN-4	It want to detect suspicious transactions	5	High	Tharugu Sree Lakshmi

			using a machine learning model, so that fraud can be identified in real-time.			
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Sprint-2	Alert System	USN-5	It want to receive an alert when fraud is detected, so that I can take immediate action.	3	High	Syed Aashifa
Sprint-3	Dashboard	USN-6	As an admin, I want to view a fraud statistics dashboard, so that I can monitor system performance.	3	Medium	Somisetty Vasanth Kumar
Sprint-3	Reporting	USN-7	It want to generate monthly fraud detection reports, so that fraud trends can be analyzed.	3	High	Tharugu Sree Lakshmi

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	4	5 Days	4 Feb 2026	9 Feb 2026	4	10 Feb 2026
Sprint-2	8	6 Days	5 Feb 2026	11 Feb 2026	8	12 Feb 2026
Sprint-3	6	6 Days	6 Feb 2026	11 Feb 2026	6	12 Feb 2026
Sprint-4	2	3 Days	7 Mar 2026	10 Feb2026	2	13 Feb 2026

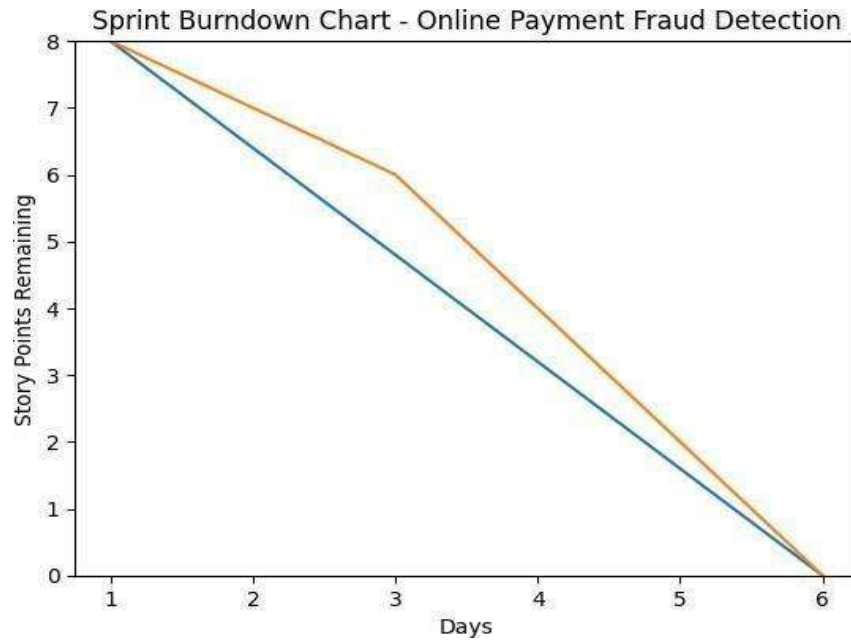
Velocity Formula

Velocity = Total Story Points / Number of Sprints

$$= 20 / 4$$

= 5 Story Points per Sprint

Burndown Chart:



CHAPTER 6

FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

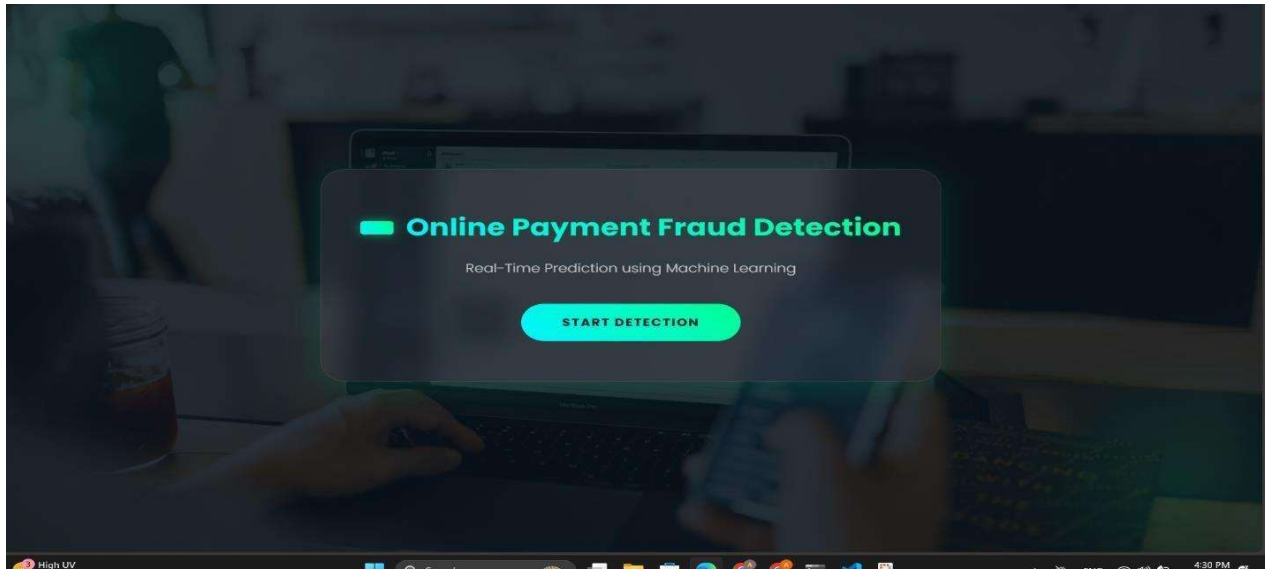
Project Development Phase Model Performance Test:

S.No.	Parameter	Values
1	Model Summary	Multiple models trained: Random Forest, Decision Tree, Extra Trees, SVC Dataset Size: 16,426 balanced samples Features Used: step, amount, balances, transaction type (one- hot encoded)
2	Accuracy	Random Forest: Training Accuracy – 100% Test Accuracy – 99.27% Decision Tree: Training Accuracy – 100% Test Accuracy – 99.48% Extra Trees: Training Accuracy – 100% Test Accuracy – 98.81% SVC: Training Accuracy – 83.58% Test Accuracy – 83.26%
3	Confusion Matrix (Best Performing Model)	True Fraud Correctly Detected – 1610 False Fraud Alerts – 6–18 Precision $\approx$ 0.99 Recall $\approx$ 0.99 F1- Score $\approx$ 0.99
4	Fine Tuning Result	Decision Tree achieved highest Test Accuracy – 99.48% Random Forest also performed strongly with 99.27% accuracy

CHAPTER 7

RESULTS

7.1 Output Screenshots



The screenshot displays a "Analyze Transaction" form on a dark purple background. The form is a light gray rounded rectangle containing several input fields and a dropdown menu. The inputs are: a text field with "5", a text field with "10000", a text field with "20000", a text field with "10000", a text field with "0", and a text field with "10000". Below these is a dropdown menu currently showing "TRANSFER". At the bottom of the form is a red button labeled "Predict".

Prediction Result

Legitim Transaction

Another Transaction

127.0.0.1:5000/predict

Summarize

Chat

Analyze Transaction

3

5000

10000

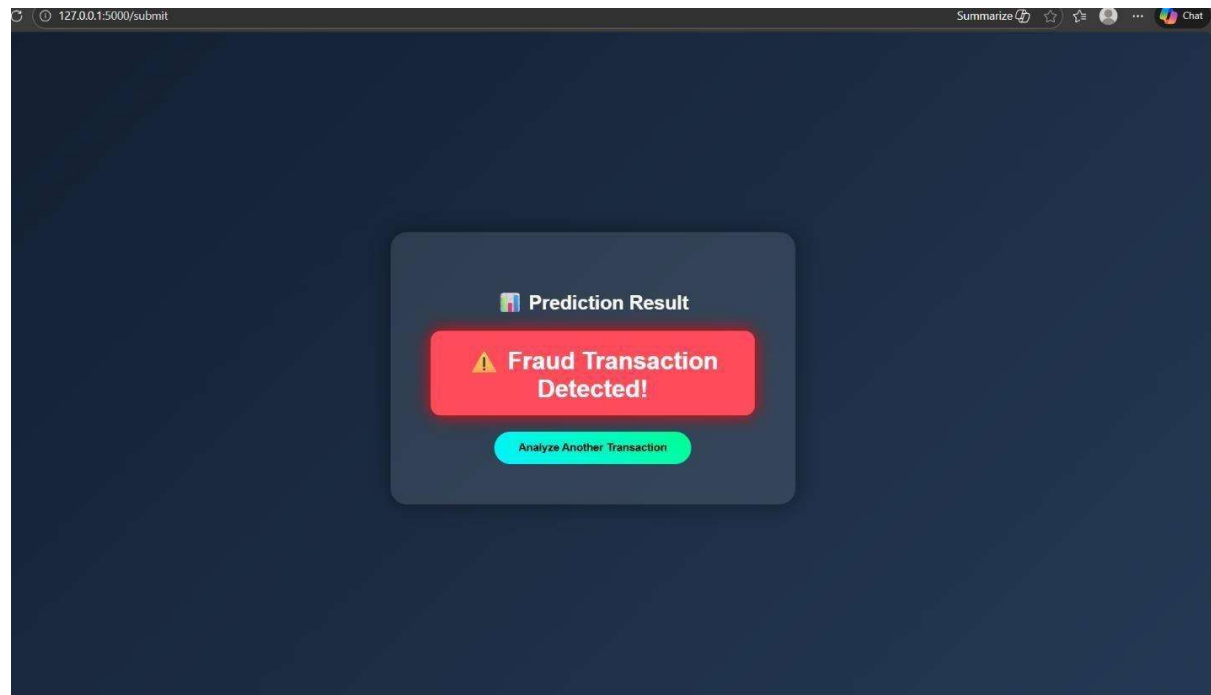
0

0

0

TRANSFER

Predict



CHAPTER 8

ADVANTAGES & DISADVANTAGES

Advantages

Real-Time Fraud Detection:

Detects suspicious transactions instantly using machine learning models.

High Accuracy:

Achieves ~99% accuracy using Decision Tree / Random Forest models.

Reduced Financial Loss:

Automatically blocks high-risk transactions before money is lost.

Improved Customer Trust:

Enhances confidence in digital payments by ensuring security.

Automated Monitoring:

Eliminates manual fraud checking and reduces human error.

Scalable Architecture:

Can handle large volumes of transactions when deployed on cloud.

Comprehensive Reporting:

Provides dashboard and fraud analytics for administrators.

Balanced Dataset Handling:

Handles imbalanced fraud datasets effectively.

Disadvantages

High Computational Cost:

Real-time ML predictions require processing power.

Risk of Overfitting:

Some models (like Decision Tree) may overfit if not properly tuned.

False Positives:

Legitimate transactions may sometimes be flagged as fraud.

Data Dependency:

Model performance depends heavily on quality and size of dataset.

Security Risk if Model Exposed:

If system APIs are not secured, attackers may exploit vulnerabilities.

Maintenance Requirement:

Requires periodic retraining to adapt to new fraud patterns.

CHAPTER 9

CONCLUSION

The Online Payment Fraud Detection System successfully addresses the growing issue of digital transaction fraud by leveraging machine learning techniques.

The system achieves high prediction accuracy (~99%) using classification models such as Random Forest and Decision Tree. It effectively analyzes transaction behavior patterns and identifies suspicious activities in real-time.

By integrating fraud detection with alert mechanisms and administrative dashboards, the system enhances financial security, reduces monetary losses, and improves user trust in digital payment systems.

Overall, the project demonstrates the practical application of machine learning in solving real-world cybersecurity challenges.

CHAPTER 10

FUTURE SCOPE

Deep Learning Integration:

Implement LSTM or Neural Networks for sequential transaction analysis.

Real-Time API Deployment:

Deploy as a scalable REST API for banks and fintech platforms.

Blockchain Integration:

Combine with blockchain for enhanced transaction transparency.

Behavioral Biometrics:

Add user behavior analysis (typing speed, device fingerprinting).

Advanced Ensemble Models:

Combine multiple ML models for improved prediction performance.

Mobile App Integration:

Provide fraud alerts directly via mobile application.

Explainable AI (XAI):

Add model interpretability to explain why a transaction was flagged.

Cloud Microservices Architecture:

Deploy using Kubernetes for large-scale banking systems.

CHAPTER 11

APPENDIX

Dataset Link:

<https://www.kaggle.com/datasets/rupakroy/online-payments-fraud-detection-dataset>

GitHub Link:

<https://github.com/AlisettySneha/Online-Payment-Fraud-Detection-using-Machine-Learning>